

K-Mean Clustering Algorithm

In this example, we use

			Cluster 1	Cluster 2	Cluster 3	
Point	X	Y	Distance from Centroid A1 (2,10)	Distance from Centroid B1 (5,8)	Distance from Centroid C1 (1,2)	Assigned Cluster
A1	2	10	0	3.605551275	8.062257748	C1
A2	2	5	5	4.242640687	3.16227766	C3
A3	8	4	8.485281374	5	7.280109889	C2
B1	5	8	3.605551275	0	7.211102551	C2
B2	7	5	7.071067812	3.605551275	6.708203932	C2
B3	6	4	7.211102551	4.123105626	5.385164807	C2
C1	1	2	8.062257748	7.211102551	0	C3
C2	4	9	2.236067977	1.414213562	7.615773106	C2

Ecludian Distance:

$$\sqrt{|x_{i1}-x_{j1}|^2 + |x_{i2}-x_{j2}|^2} \dots$$

Cluster 1 Mean1 2

Cluster 2 Mean1 6

Cluster 3 Mean1 1.5

Cluster 1 Mean2 10

Cluster 2 Mean2 6

Cluster 3 Mean2 3.5

			Cluster 1	Cluster 2	Cluster 3	
Point	X	Y	Distance from Centroid A1 (2,10)	Distance from Centroid B1 (6,6)	Distance from Centroid C1 (1.5,3.5)	Assigned Cluster
A1	2	10	0	5.656854249	6.519202405	C1
A2	2	5	5	4.123105626	1.58113883	C3
A3	8	4	8.485281374	2.828427125	6.519202405	C2
B1	5	8	3.605551275	2.236067977	5.700877125	C2
B2	7	5	7.071067812	1.414213562	5.700877125	C2
B3	6	4	7.211102551	2	4.527692569	C2
C1	1	2	8.062257748	6.403124237	1.58113883	C3
C2	4	9	2.236067977	3.605551275	6.041522987	C2

Manhattan Distance

$$|x_{i1}-x_{j1}| + |x_{i2}-x_{j2}| \dots$$

Supremum Distance

$$\text{Max} (|x_{i1}-x_{j1}| + |x_{i2}-x_{j2}| \dots$$

